

Deep Learning

Vazgen Mikayelyan

May 22, 2021



Outline

- 1 Moving Average
- 2 Batch Normalization
- 3 Other Optimizers
- 4 Data Augmentation
- 5 What is Convolutional Neural Network?

Simple Moving Average

Definition 1

Simple moving average of the given data is the arithmetic mean of the previous k data.

Simple Moving Average

Definition 1

Simple moving average of the given data is the arithmetic mean of the previous k data.

If you have the data $x_1, x_2, \dots$, then its simple moving average will be the following

$$\mu_n = \frac{x_{n-k+1} + \dots + x_n}{k}, n = k, k+1, \dots$$

Cumulative Moving Average

Definition 2

Cumulative moving average of the given data is the arithmetic mean of the all previous data up to the current time.

Cumulative Moving Average

Definition 2

Cumulative moving average of the given data is the arithmetic mean of the all previous data up to the current time.

If you have the data $x_1, x_2, \dots$, then its cumulative moving average will be the following

$$\mu_n = \frac{x_1 + x_2 + \dots + x_n}{n}, n = 1, 2, \dots$$

Cumulative Moving Average

Definition 2

Cumulative moving average of the given data is the arithmetic mean of the all previous data up to the current time.

If you have the data $x_1, x_2, \dots$, then its cumulative moving average will be the following

$$\mu_n = \frac{x_1 + x_2 + \dots + x_n}{n}, n = 1, 2, \dots$$

Note that

$$\begin{aligned}\mu_n &= \frac{(x_1 + x_2 + \dots + x_{n-1}) + x_n}{n} = \frac{(n-1)\mu_{n-1} + x_n}{n} \\ &= \left(1 - \frac{1}{n}\right) \mu_{n-1} + \frac{1}{n} x_n.\end{aligned}$$

Exponential Moving Average

If you have the data $x_1, x_2, \dots$, then its exponential moving average will be the following

$$\begin{aligned}\mu_1 &= x_1, \\ \mu_n &= \alpha\mu_{n-1} + (1 - \alpha)x_n, \quad n \geq 2,\end{aligned}$$

Exponential Moving Average

If you have the data $x_1, x_2, \dots$, then its exponential moving average will be the following

$$\mu_1 = x_1,$$

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n, \quad n \geq 2,$$

where $\alpha \in (0, 1)$. Note that

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n = \alpha (\alpha \mu_{n-2} + (1 - \alpha) x_{n-1}) + (1 - \alpha) x_n$$

Exponential Moving Average

If you have the data $x_1, x_2, \dots$, then its exponential moving average will be the following

$$\mu_1 = x_1,$$

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n, \quad n \geq 2,$$

where $\alpha \in (0, 1)$. Note that

$$\begin{aligned} \mu_n &= \alpha \mu_{n-1} + (1 - \alpha) x_n = \alpha (\alpha \mu_{n-2} + (1 - \alpha) x_{n-1}) + (1 - \alpha) x_n \\ &= \alpha^2 \mu_{n-2} + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n \end{aligned}$$

Exponential Moving Average

If you have the data $x_1, x_2, \dots$, then its exponential moving average will be the following

$$\mu_1 = x_1,$$

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n, \quad n \geq 2,$$

where $\alpha \in (0, 1)$. Note that

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n = \alpha (\alpha \mu_{n-2} + (1 - \alpha) x_{n-1}) + (1 - \alpha) x_n$$

$$= \alpha^2 \mu_{n-2} + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n$$

$$= \alpha^3 \mu_{n-3} + (1 - \alpha) \alpha^2 x_{n-2} + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n$$

Exponential Moving Average

If you have the data $x_1, x_2, \dots$, then its exponential moving average will be the following

$$\mu_1 = x_1,$$

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n, \quad n \geq 2,$$

where $\alpha \in (0, 1)$. Note that

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n = \alpha (\alpha \mu_{n-2} + (1 - \alpha) x_{n-1}) + (1 - \alpha) x_n$$

$$= \alpha^2 \mu_{n-2} + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n$$

$$= \alpha^3 \mu_{n-3} + (1 - \alpha) \alpha^2 x_{n-2} + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n$$

$$= \alpha^{n-1} \mu_1 + (1 - \alpha) \alpha^{n-2} x_2 + \dots + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n.$$

Exponential Moving Average

If you have the data $x_1, x_2, \dots$, then its exponential moving average will be the following

$$\mu_1 = x_1,$$

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n, \quad n \geq 2,$$

where $\alpha \in (0, 1)$. Note that

$$\mu_n = \alpha \mu_{n-1} + (1 - \alpha) x_n = \alpha (\alpha \mu_{n-2} + (1 - \alpha) x_{n-1}) + (1 - \alpha) x_n$$

$$= \alpha^2 \mu_{n-2} + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n$$

$$= \alpha^3 \mu_{n-3} + (1 - \alpha) \alpha^2 x_{n-2} + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n$$

$$= \alpha^{n-1} \mu_1 + (1 - \alpha) \alpha^{n-2} x_2 + \dots + (1 - \alpha) \alpha x_{n-1} + (1 - \alpha) x_n.$$

It easy to see that sum of the coefficients is equal to 1.

Outline

- 1 Moving Average
- 2 Batch Normalization
- 3 Other Optimizers
- 4 Data Augmentation
- 5 What is Convolutional Neural Network?

Batch Normalization

- Problem:
 - The distribution of each layer's input changes during training.

Batch Normalization

- Problem:
 - The distribution of each layer's input changes during training.
- Solution:
 - Fix the distribution of inputs into subnetwork.

Batch Normalization

- Problem:
 - The distribution of each layer's input changes during training.
- Solution:
 - Fix the distribution of inputs into subnetwork.
- Effects:
 - Improve accuracy.
 - Faster learning.
 - Availability of high learning rates.

Batch Normalization

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_1 \dots x_m\}$;

Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$

$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$

- 1 Can we do stochastic gradient descent in this case?

Questions

- 1 Can we do stochastic gradient descent in this case?
- 2 What to do during the test?

- 1 Can we do stochastic gradient descent in this case?
- 2 What to do during the test?
- 3 What about biases?

Outline

- 1 Moving Average
- 2 Batch Normalization
- 3 Other Optimizers**
- 4 Data Augmentation
- 5 What is Convolutional Neural Network?

Gradient Descent with Momentum

Let $L(w)$ be a loss function that we want to minimize. The algorithm gradient descent with momentum is the following

$$v_0 = 0,$$

$$v_t = \beta v_{t-1} + (1 - \beta) \nabla L(w_t),$$

$$w_{t+1} = w_t - \alpha v_t,$$

where α is the learning rate and $\beta \in [0, 1)$ is the parameter of exponential moving average.

Gradient Descent with Momentum



Image 2: SGD without momentum



Image 3: SGD with momentum

Let $L(w)$ be a loss function that we want to minimize. The algorithm RMSprop is the following

$$v_0 = 0,$$

$$v_t = \beta v_{t-1} + (1 - \beta) (\nabla L(w_t))^2,$$

$$w_{t+1} = w_t - \alpha \frac{\nabla L(w_t)}{\sqrt{v_t} + \varepsilon},$$

where α is the learning rate and $\beta \in [0, 1)$ is the parameter of exponential moving average.

ADAM

Let $L(w)$ be a loss function that we want to minimize. The algorithm ADAM is the following

$$\begin{aligned}m_0 &= 0, v_0 = 0, \\m_t &= \beta_1 m_{t-1} + (1 - \beta_1) \nabla L(w_{t-1}), \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t},\end{aligned}$$

Let $L(w)$ be a loss function that we want to minimize. The algorithm ADAM is the following

$$m_0 = 0, v_0 = 0,$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla L(w_{t-1}),$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t},$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla L(w_{t-1}))^2,$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t},$$

Let $L(w)$ be a loss function that we want to minimize. The algorithm ADAM is the following

$$m_0 = 0, v_0 = 0,$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla L(w_{t-1}),$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t},$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla L(w_{t-1}))^2,$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t},$$

$$w_t = w_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon},$$

where α is the learning rate and $\beta_1, \beta_2 \in [0, 1)$ are the parameters of exponential moving averages.

Outline

- 1 Moving Average
- 2 Batch Normalization
- 3 Other Optimizers
- 4 Data Augmentation**
- 5 What is Convolutional Neural Network?

- Images:

Data Augmentation

- Images:
 - horizontal flips,

Data Augmentation

- Images:
 - horizontal flips,
 - crops,

- Images:
 - horizontal flips,
 - crops,
 - change contrast of colours,

Data Augmentation

- Images:
 - horizontal flips,
 - crops,
 - change contrast of colours,
 - add noise.
- Audios:

Data Augmentation

- Images:
 - horizontal flips,
 - crops,
 - change contrast of colours,
 - add noise.
- Audios:
 - change speed,

Data Augmentation

- Images:

- horizontal flips,
- crops,
- change contrast of colours,
- add noise.

- Audios:

- change speed,
- change pitch,

Data Augmentation

- Images:

- horizontal flips,
- crops,
- change contrast of colours,
- add noise.

- Audios:

- change speed,
- change pitch,
- add noise.

Outline

- 1 Moving Average
- 2 Batch Normalization
- 3 Other Optimizers
- 4 Data Augmentation
- 5 What is Convolutional Neural Network?

What is convolution?

Definition 3

Convolution of the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as the integral of the product of the two functions after one is reversed and shifted:

$$(f * g)(t) =: \int_{-\infty}^{+\infty} f(x) g(t - x) dx.$$

What is convolution?

Definition 3

Convolution of the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as the integral of the product of the two functions after one is reversed and shifted:

$$(f * g)(t) =: \int_{-\infty}^{+\infty} f(x) g(t - x) dx.$$

It easy to see that $f * g = g * f$.

What is convolution?

Definition 3

Convolution of the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as the integral of the product of the two functions after one is reversed and shifted:

$$(f * g)(t) =: \int_{-\infty}^{+\infty} f(x) g(t - x) dx.$$

It easy to see that $f * g = g * f$.

Definition 4

Convolution of the sequences of real numbers $\{f_n\}_{n=-\infty}^{+\infty}$, $\{g_n\}_{n=-\infty}^{+\infty}$ is the following sequence:

$$z_n =: \sum_{k=-\infty}^{+\infty} f_k g_{n-k}.$$